AI Playbook and Toolkit for Public Health Departments

Synthetic Data for Public Health AI

ANL 320

Synthetic data are artificially generated data that are designed to resemble real data without directly copying every record from a real person. In public health, synthetic data can support training, software testing, demonstrations, model prototyping, and vendor evaluation when real data are too sensitive or difficult to share. This module helps learners understand both the promise and limitations of synthetic data, including privacy risk, analytic utility, representativeness, and validation.

Level	intermediate/practitioner
Primary track	Analytics and Modeling Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of synthetic data for public health ai in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a synthetic dataset assessment that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

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Why This Topic Matters for Public Health AI

Synthetic Data for Public Health AI matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

Synthetic data are generated data designed to preserve selected patterns of a source dataset while reducing reliance on real individual-level records. The value of synthetic data depends on the purpose. A dataset that is useful for training staff on a dashboard may not be valid for evaluating a predictive model or testing equity performance.

Privacy risk does not disappear simply because data are synthetic. If synthetic data are too close to the source data, they may leak information about real individuals or rare cases. If they are too different from the source data, they may be safe but not useful. Public health agencies must evaluate both privacy protection and data utility.

Data utility means the synthetic data are useful for the intended purpose. Utility may involve preserving distributions, relationships between fields, missingness patterns, geographic variation, or time trends. The intended use should be documented before the synthetic data are created or accepted from a vendor.

Public Health Example

A health department may want synthetic eCR data to train disease investigators and test an AI case summary tool without exposing real case narratives. The synthetic data should include realistic combinations of symptoms, lab results, demographics, dates, and missing fields, but should not reproduce unusual cases in ways that could identify real people. The agency should document that the data are intended for training and testing, not for estimating disease burden or measuring model performance in production.

Technical Deep Dive

Synthetic data development begins by defining the intended use. Training data for staff exercises can be less statistically faithful than data used for system testing. Model evaluation data require more rigorous review because the synthetic data must preserve patterns relevant to the model task. Public health teams should define whether synthetic data will be used for demonstration, software testing, integration testing, training, simulation, or model development.

Agencies should evaluate synthetic data for privacy and utility. Privacy review may examine whether rare combinations, dates, locations, or narratives are too similar to real records. Utility review may compare distributions, relationships, missingness, and edge cases against expected patterns. Both reviews should be documented because a dataset can be safe but not useful, or useful but not safe enough for sharing.

Synthetic data are also helpful for vendor evaluation. Agencies can provide vendors with realistic data structures without sharing protected data during early discovery. However, vendors should not use synthetic data performance as proof that a model will work on real public health data. Production readiness still requires validation against appropriate real-world data under approved governance controls.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a synthetic dataset assessment for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed synthetic dataset assessment. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

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Knowledge Check

- 1. What is the best reason to address synthetic data for public health ai before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Privacy Framework and differential privacy resources
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies